Module 12: Wireless Networking Fundamentals

Intro to Networks

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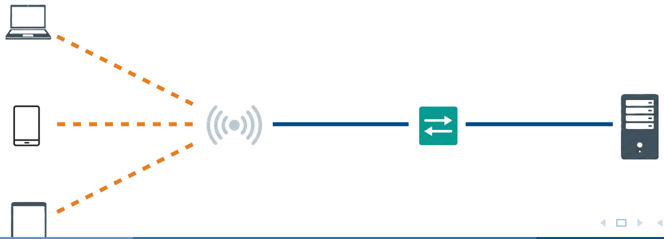


Outline I

Review: Security Zones and Physical Controls

Key Module 11 Ideas

- **Security zones** separate traffic by trust level (Internet, DMZ, Internal).
- **IoT devices** have long lifecycles and poor patch support—they need isolated VLANs.
- **Physical security** (locks, cameras, mantraps) is the last line of defense.
- **Defense in depth** means layering many controls so no single failure is catastrophic.

Where Module 12 Takes This

Module 11 assumed a *wired* boundary you could physically protect. Module 12 tears down the wall: Wi-Fi broadcasts your network into open air, visible to anyone nearby. Understanding **how radio waves work**, **how standards evolved**, and **how to secure an airwave** is the focus.

Learning Outcomes I

After completing this module, you will be able to:

- Explain how **radio waves** carry data and how frequency, wavelength, and amplitude affect wireless performance.
- Describe the trade-offs between the **2.4 GHz**, **5 GHz**, and **6 GHz** frequency bands.
- Explain why only channels **1, 6, and 11** are used in 2.4 GHz, and what **co-channel** vs. **adjacent-channel interference** means.
- Describe how **CSMA/CA** prevents collisions and how **RTS/CTS** solves the hidden node problem.
- Compare 802.11 standards (b, a, g, n, ac, ax, be) and identify the key technology added by each generation.
- Explain the difference between **autonomous (fat) APs** and **lightweight (thin) APs** managed by a **WLC**.
- Design a basic enterprise wireless layout using appropriate **site survey** methods and antenna types.
- Compare **WEP**, **WPA**, **WPA2**, and **WPA3** and explain why older protocols were replaced.
- Distinguish **Personal (PSK)** from **Enterprise (802.1X)** authentication and explain when each is appropriate.

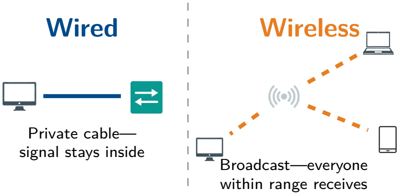
Learning Outcomes II

- Identify wireless threats—**rogue APs**, **evil twin attacks**—and describe defenses.
- Interpret **RSSI** and **SNR** values and troubleshoot common wireless issues.

Section 1: How Radio Waves Carry Data

What Makes Wireless Different?

- In a **wired** network, your data travels as electrical signals or light inside a *private* cable—only your device and the switch see it.
- In a **wireless** network, your data is broadcast as radio waves into open air—*everyone within range* can receive it.
- This changes everything: security, performance, and interference all work differently when the medium is invisible and shared.



Analogy

A wired network is like whispering through a private tube. A wireless network is like shouting across a crowded room—everyone hears you, and you can only talk when no one else is talking.

What Is a Radio Wave? Frequency, Wavelength, Amplitude

- A **radio wave** is an electromagnetic wave—the same physical phenomenon as visible light, just at a much lower frequency.
- **Frequency (f):** How many wave cycles occur per second. Measured in Hertz (Hz). Wi-Fi uses *billions* of cycles per second—the GHz range.
- **Wavelength (λ):** The physical distance between two peaks. Since the speed of light is constant ($c = \lambda f$), higher frequency = shorter wavelength.
- **Amplitude:** The height of the wave—how much energy it carries. Stronger signal = higher amplitude.

Why This Matters in Practice

Lower frequency (2.4 GHz): Longer wavelength, passes through walls easily, travels farther—but the band is crowded.

Higher frequency (5 GHz): Shorter wavelength, faster data rates, but absorbed by walls and human bodies more readily—shorter range.

Ocean Wave Analogy

Think of frequency as how rapidly waves arrive at the beach. Low frequency = slow gentle rollers that travel far. High frequency = rapid choppy waves that die out quickly.

Half-Duplex: The Walkie-Talkie Constraint

- A wireless radio uses the **same antenna** to both transmit and receive.
- It physically cannot do both at the same time—transmitting would deafen its own receiver.
- This means **all Wi-Fi is half-duplex**: only one device on a channel can transmit at any instant.
- **Modulation:** To encode data, the AP rapidly changes properties of the carrier wave—altering its phase or amplitude—to represent 1s and 0s.
- **QAM (Quadrature Amplitude Modulation):** Modern Wi-Fi changes both phase *and* amplitude simultaneously. 1024-QAM encodes 10 bits per symbol.

Walkie-Talkie Analogy

Wireless works exactly like a walkie-talkie:

"Can you hear me? Over."
"Yes, loud and clear. Over."

Both sides share one frequency. If two people transmit simultaneously, signals collide and neither message gets through—hence the need for *collision avoidance*.

Contrast with Wired Ethernet

A switched Ethernet port is *full-duplex*: it can send and receive simultaneously because the Tx and Rx wires are physically separate.

Frequency Bands: 2.4 GHz, 5 GHz, and 6 GHz

Property	2.4 GHz	5 GHz	6 GHz
Range	Excellent (far, penetrates walls)	Moderate (shorter range)	Short (mostly indoor)
Speed	Limited (few channels, congested)	High (many wide channels)	Very High (fresh spectrum)
Congestion	Very crowded (microwaves, Bluetooth, neighbors)	Relatively clean	Uncrowded (new!)
Non-overlapping ch.	Only **3** (1, 6, 11)	Up to **24** at 20 MHz	Up to **59** at 20 MHz
Standard	802.11b/g/n/ax	802.11a/n/ac/ax	802.11ax (6E), 802.11be

Rule of Thumb

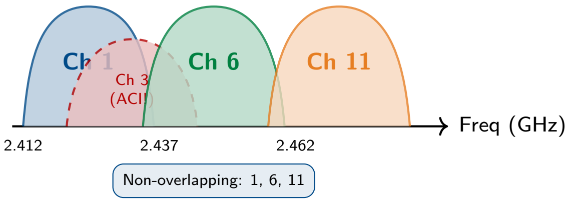
Choose **2.4 GHz** when range matters most (large homes, legacy IoT). Choose **5 or 6 GHz** when speed matters most (video, large files).

The 6 GHz Greenfield Advantage

The 6 GHz band (Wi-Fi 6E and Wi-Fi 7) is *brand new* spectrum with zero legacy devices. It **requires WPA3**—no open networks or WPA2 allowed.

Channels: Dividing the Spectrum into Lanes

- The 2.4 GHz band is about 72 MHz wide. The US defines 11 numbered **channels**, each 22 MHz wide—but spaced only **5 MHz apart**.
- **The Math Problem:** Because channels are wider than their spacing, most physically *overlap* in frequency.
- **Co-Channel Interference (CCI):** Two APs on the *same* channel. They politely take turns (CSMA/CA). Manageable.
- **Adjacent-Channel Interference (ACI):** Two APs on *overlapping but different* channels. They cannot coordinate—RF static. Far worse than CCI.
- In the US, only channels **1, 6, and 11** are far enough apart to be non-overlapping.



Key Rule

Never assign a 2.4 GHz AP to channels 2–5, 7–10, or 12. Use only **1, 6, or 11**.

CSMA/CA: "Listen Before You Talk"

- Wired Ethernet used CSMA/**CD** (Collision *Detection*): transmit, detect a collision after the fact, and retransmit. This works because wired devices *can* hear a collision while transmitting.
- Wi-Fi cannot detect collisions while transmitting (half-duplex), so it uses CSMA/**CA** (Collision *Avoidance*).
- **Step 1 – Listen:** Before transmitting, the device checks if the channel is idle (**Clear Channel Assessment**).
- **Step 2 – Wait:** If busy, the device sets a random **backoff timer** and waits.
- **Step 3 – Transmit:** When the timer expires and the channel is still idle, the device sends.
- **Step 4 – ACK:** The receiver confirms receipt. No ACK = retransmit.

Classroom Analogy

CSMA/CA works like a polite classroom: before speaking, you *listen* to ensure no one else is talking. If someone is, you wait a random amount of time before trying again. The random wait prevents everyone from retrying at the same moment.

Why Backoff is Random

If all devices waited the *same* amount of time, they would all retry simultaneously and collide again. A random timer means they retry at different moments, letting one get through first.

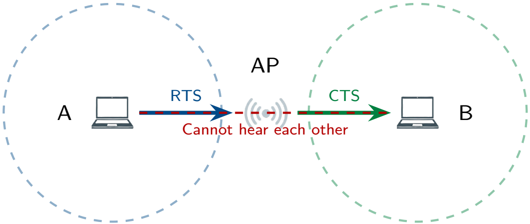
The Hidden Node Problem and RTS/CTS

The Hidden Node Problem

Laptop A and Laptop B can both reach the AP, but they are on opposite sides of the building and *cannot hear each other*. CSMA/CA fails: each thinks the channel is idle and transmits simultaneously, causing a collision at the AP.

The Fix: RTS/CTS

1. Laptop A sends a tiny **RTS** to the AP.
2. The AP broadcasts a **CTS** to *all* devices.
3. All devices set a **NAV** timer to stay silent.
4. Laptop A transmits data safely.



RTS/CTS breaks the hidden-node deadlock

Section 2: Wi-Fi Standards

The 802.11 Family: A 25-Year Journey

- The **IEEE 802.11** standard defines the physical (PHY) and media-access (MAC) layers for wireless LANs. Each new generation added a key technology to improve speed, range, or efficiency.
- The **Wi-Fi Alliance** introduced friendly names (Wi-Fi 4, 5, 6...) for consumer marketing.

Standard	Wi-Fi Name	Band(s)	Max Speed	Key Innovation
802.11b (1999)	Wi-Fi 1	2.4 GHz	11 Mbps	**DSSS** – reliable, slow; started the Wi-Fi revolution
802.11a (1999)	Wi-Fi 2	5 GHz	54 Mbps	**OFDM** – splits signal into sub-carriers; much faster
802.11g (2003)	Wi-Fi 3	2.4 GHz	54 Mbps	Brought OFDM's speed to the 2.4 GHz band
802.11n (2009)	Wi-Fi 4	2.4 & 5 GHz	600 Mbps	**MIMO** – multiple antennas, multiple data streams
802.11ac (2013)	Wi-Fi 5	5 GHz only	6.9 Gbps	**MU-MIMO** (downlink), 160 MHz channel bonding
802.11ax (2019)	Wi-Fi 6/6E	2.4, 5, & 6 GHz	9.6 Gbps	**OFDMA** – serves many clients per transmission
802.11be (2024+)	Wi-Fi 7	2.4, 5, & 6 GHz	46 Gbps	**MLO** – simultaneous multi-band links

Early Standards: 802.11b, a, and g

802.11b — The Standard That Started It All

Used **DSSS (Direct Sequence Spread Spectrum)**: data is spread across a wide swath of the 2.4 GHz band simultaneously. Very resistant to interference, but tops out at just **11 Mbps**. Its simplicity made Wi-Fi affordable for consumers in the early 2000s.

802.11a — The Fast but Forgotten Standard

Operated only on 5 GHz using **OFDM (Orthogonal Frequency Division Multiplexing)**: the channel is split into many narrow sub-carriers transmitted in parallel, reaching **54 Mbps**. Great speed, but poor wall penetration limited its popularity.

802.11g — The Grand Compromise

Took 802.11a's OFDM speed and brought it to the popular 2.4 GHz band: **54 Mbps** with good range. Backwards-compatible with 802.11b devices.

The Backwards-Compatibility Trap

Supporting 802.11b *required* the AP to send special slow protection frames (at 1–11 Mbps) so old devices would not interrupt fast clients. This overhead ate up airtime for everyone. **Best practice on modern networks: disable legacy (802.11b) data rates entirely.**

802.11n (Wi-Fi 4): MIMO Turns Multipath into an Advantage

- **The old problem:** A signal bouncing off walls arrives at the receiver from multiple paths, slightly delayed. These **multipath echoes** interfere with each other and corrupt the data.
- **The MIMO insight:** What if those separate paths each carried a *different* stream of data? 802.11n introduced **MIMO (Multiple Input, Multiple Output)**: an AP with 3 antennas can transmit 3 independent spatial streams to a 3-antenna client.
- Result: the same channel now carries 3× the data. 802.11n reached **600 Mbps** on 5 GHz with 4 streams.

Highway Lane Analogy

Before MIMO, a channel was a one-lane road. MIMO adds more lanes to the same road—the speed limit (frequency) hasn't changed, but multiple cars travel simultaneously.

Reading an Antenna Spec

Access points are listed as **Tx×Rx:Streams** (e.g., **3×3:3**).

- **3×3:3** = 3 transmit, 3 receive, 3 spatial streams.
- The client must also support 3 streams to benefit.
- A 3×3:3 AP talking to a 1×1:1 phone gives only 1 stream.

802.11ac (Wi-Fi 5): Wider Channels and Multi-User MIMO

- **5 GHz only:** 802.11ac moved entirely to 5 GHz to gain access to the many uncongested channels there.
- **Channel Bonding:** Instead of a single 20 MHz channel, 802.11ac bonds multiple channels together—40, 80, or even 160 MHz—creating a much wider "highway." An 80 MHz channel carries roughly 4$\times$ as much data as a 20 MHz channel.
- **MU-MIMO (Multi-User MIMO):** 802.11n served one client at a time. 802.11ac can transmit to *multiple clients simultaneously* by mathematically steering separate beams.

The Downlink-Only Limitation

In 802.11ac, MU-MIMO only works on **downlink** (AP to client). Clients still take turns sending *up* to the AP. This was fixed in Wi-Fi 6.

256-QAM

802.11ac introduced **256-QAM**: each RF symbol now encodes 8 bits (up from 6 bits in 64-QAM). More bits per symbol = higher throughput—but only when the client is close enough for a clean, high-SNR signal.

802.11ax (Wi-Fi 6) and 802.11be (Wi-Fi 7)

Wi-Fi 6 — Designed for Crowded Networks

- Wi-Fi 5 was fast in the lab; Wi-Fi 6 is fast *at a stadium concert*.
- **OFDMA**: The AP subdivides a single 20 MHz channel into smaller **Resource Units (RUs)** and sends data to *many clients in one transmission*—like packing multiple letters into one envelope.
- **BSS Coloring:** APs add a "color" tag so nearby APs on the same channel can identify and ignore each other's traffic, reducing unnecessary backoffs.
- **Target Wake Time (TWT):** IoT devices agree on sleep schedules with the AP, saving battery.

Wi-Fi 7 — Ultra-High Throughput

- **MLO (Multi-Link Operation):** A device connects to two bands *simultaneously* (e.g., 5 + 6 GHz), aggregating their bandwidth. If one band gets congested, traffic shifts to the other instantly.
- **4096-QAM:** 12 bits per symbol—requires near-perfect signal conditions.
- **320 MHz channels** in 6 GHz for massive throughput.
- Targets **46 Gbps**—designed to replace wired cables for VR and 8K video.

Case Study: Dustin's Airtime Anomaly

▶ Case Study: The Legacy Device Drag

Dustin Henderson, Senior RF Engineer at Hawkins National Lab, is frustrated. The lab just spent $50,000 upgrading the research wing to brand-new **Wi-Fi 6 (802.11ax)** access points. On paper the network should fly.

Instead, researchers complain the network *periodically crawls to a halt*. Dustin pulls a packet capture from the WLC and finds high channel utilization and massive retries—then spots the culprit: a **single barcode scanner** running 802.11b at 11 Mbps is associating to the 2.4 GHz network during inventory checks.

Review Questions

1. What frequency band and maximum speed does the 802.11b barcode scanner operate on?
2. Wi-Fi is shared and half-duplex. Why would *one* slow device degrade performance for *all* other, faster devices?
3. What specific configuration change should Dustin make to resolve this permanently?

Case Study Solution: Dustin's Airtime Anomaly

✓ Solution: The Legacy Device Drag

1. The barcode scanner uses the **2.4 GHz band** at a maximum of **11 Mbps**.
2. **The Slow-Client / Airtime Problem:** Wi-Fi channels are time-shared. Because the scanner transmits at 11 Mbps, it occupies the channel for *much longer* per byte than a Wi-Fi 6 laptop at 500+ Mbps. Every millisecond the scanner is transmitting, every other device must wait. The scanner monopolizes a disproportionate share of *airtime* just to move a tiny payload.
3. **Disable Legacy Data Rates:** Dustin should configure the Wireless LAN Controller to disable 802.11b mandatory rates (1, 2, 5.5, and 11 Mbps). The AP will then refuse to associate with any 802.11b device, protecting airtime for modern clients.

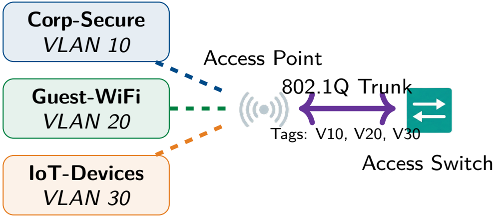
Key Takeaway

One legacy device on a shared Wi-Fi channel is like one car driving 15 mph on a freeway—everyone behind it is stuck. Disabling legacy rates forces the scanner to be replaced, but protects the entire network.

Section 3: Enterprise Wireless Design

Logical Design: SSIDs, VLANs, and Trunking

- An **SSID (Service Set Identifier)** is the network name clients see when searching for Wi-Fi.
- One physical AP typically *broadcasts multiple SSIDs* at once—for example: `Corp-Secure`, `Guest-WiFi`, and `IoT-Devices`.
- Each SSID is logically mapped to a different **VLAN**, keeping traffic separated just like separate physical networks.
- Because multiple VLANs must travel over a single cable from the AP to the switch, that link must be an **802.1Q trunk**—it tags each frame with the VLAN ID.



Autonomous vs. Lightweight APs: Scale Changes Everything

Autonomous (Fat) APs

- Fully self-contained: each AP manages its own authentication, security, QoS, and SSID settings.
- Works great for a home router or a single-AP coffee shop.
- **The scaling problem:** Imagine an enterprise with **500 APs**. Changing the Wi-Fi password—or updating firmware—means logging into 500 separate devices. One misconfigured AP breaks policy. Nightmare.

Lightweight (Thin) APs

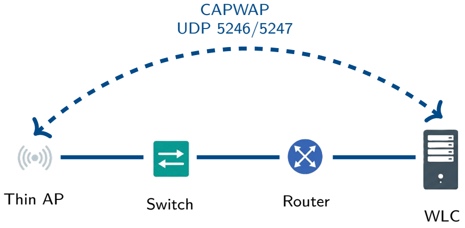
- The AP is reduced to a simple radio: it beacons, transmits, and forwards frames.
- All "intelligence" (auth, roaming, channel selection, policy) lives in a centralized **Wireless LAN Controller (WLC)**.
- **The benefit:** Change a password on the WLC once—all 500 APs update instantly. Add a new SSID—all APs broadcast it within seconds.
- Enterprise networks almost always use this architecture.

The WLC is a Critical Infrastructure Device

If the WLC fails and thin APs cannot reach it, wireless service for the entire site goes down. Always deploy WLCs in redundant pairs.

The WLC and CAPWAP: How Thin APs Phone Home

- Thin APs communicate with the WLC using **CAPWAP (Control and Provisioning of Wireless Access Points)**, over UDP ports **5246** (control) and **5247** (data).
- **Split-MAC Architecture:**
  - **AP:** Time-sensitive RF tasks—beacons, ACKs, encryption.
  - **WLC:** Policy tasks—authentication, roaming decisions, channel allocation.
- **Data Tunneling:** In most designs, user traffic is encapsulated inside a CAPWAP tunnel and sent to the WLC before being placed on the wired LAN, giving the WLC full visibility and control.



Design Implication

Since all traffic flows through the WLC, it must sit at a high-bandwidth point—typically in the data center or network core.

The Site Survey: Measure Before You Mount

- A **site survey** is the process of systematically planning AP placement to achieve adequate coverage, capacity, and minimal interference *before* you start drilling holes in the ceiling.

Predictive Survey

Done *before* installation, using CAD floor plans in simulation software (e.g., Ekahau). You draw walls and assign each material an attenuation value (thick concrete vs. drywall). The software calculates predicted coverage zones so you can position APs on paper first.

Active Survey

Done *during or after* installation. An engineer walks the building with a laptop, actively associating to test APs ("AP on a stick") and measuring real-world RSSI, throughput, and packet loss. Produces a heat map of actual coverage.

Passive Survey

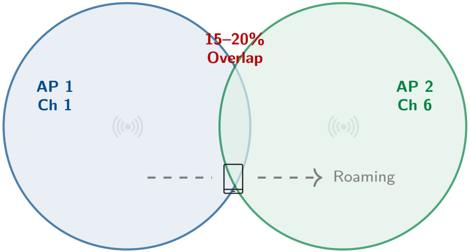
Walk the site with a spectrum analyzer set only to *listen*—never transmit. Used to find existing RF noise, unauthorized APs, and non-Wi-Fi interference (microwaves, Bluetooth) before planning begins.

Best Practice: Survey Order

Passive → Predictive → Install → Active. A passive survey first reveals interference you must design around.

Antenna Types and Coverage Cell Design

- The antenna determines the *shape* of the RF coverage zone, not just its size.
- **Omnidirectional (Dipole):** Radiates equally in all horizontal directions—a donut shape. Ideal for ceiling mounts in open-plan offices.
- **Directional (Patch / Yagi):** Focuses RF in one direction, increasing reach that way but reducing it elsewhere. Ideal for:
  - Long hallways
  - High warehouse ceilings (aimed down)
  - Bridging two buildings
- **Roaming Overlap:** Adjacent APs need **15–20% coverage overlap** so clients smoothly hand off as the user walks.



Channel Rule for Adjacent Cells

Adjacent APs *must* use non-overlapping channels. If AP1 is on Ch 1, then AP2 must be on Ch 6—never Ch 2 or 3.

Case Study: Robin's Starcourt Site Survey

▶ Case Study: The Food Court Dilemma

Robin Buckley is the network engineer designing Wi-Fi coverage for Starcourt Mall. She has two specific problems:

Problem 1 – Food Court Density: The food court hosts 200+ simultaneous smartphones during the lunch rush. A single AP in the center keeps dropping connections and running slow.

Problem 2 – Signal Bleed: The back-office SSID is bleeding out through the glass walls into the parking lot. Customers are accidentally associating to the corporate network from their cars.

Review Questions

1. For the dense food court, should Robin use a *few high-powered* APs or *many low-powered* APs? Why?
2. What specific type of antenna should Robin use near the glass walls to stop the signal from escaping into the parking lot?

Case Study Solution: Robin's Starcourt Site Survey

✓ Solution: The Food Court Dilemma

1. **Many Low-Powered APs (Microcells):** A single high-powered AP creates one giant coverage cell—all 200+ devices fight for airtime on one channel. Deploying many low-powered APs creates small cells that allow *channel reuse*: AP1 on Ch 1, AP2 on Ch 6, AP3 on Ch 11, AP4 back on Ch 1, etc. This dramatically increases total capacity because each AP serves far fewer clients.
2. **Directional (Patch) Antennas:** Robin should mount patch antennas on the interior side of the perimeter walls, *aimed inward*. A patch antenna concentrates its beam in one direction—facing into the mall. Standard omnidirectional APs broadcast a 360° donut, sending half the energy directly through the glass into the parking lot.

Key Takeaway

Good wireless design is about precisely *shaping* the RF cell—not blasting the strongest signal possible. Coverage, capacity, and containment must all be considered.

Section 4: Wireless Security

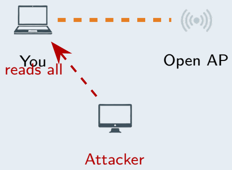
Why Wireless Encryption is Non-Negotiable

- A wired network requires physical access—an attacker must plug a cable in.
- A wireless network broadcasts every frame into the air. Anyone with a Wi-Fi adapter and free software like Wireshark can *capture everything* transmitted on an open (unencrypted) network.
- On an open Wi-Fi network, captured traffic includes: web sessions, passwords, emails, and file transfers—all in plaintext.
- **Encryption** converts data into an unreadable form so even if an attacker captures every frame, they cannot read the contents without the key.

Try This Right Now

Next time you connect to an open Wi-Fi hotspot (coffee shop, hotel), every HTTP packet you send can be captured by anyone else in the room. This is why **HTTPS** (TLS) and **VPNs** are especially critical on untrusted Wi-Fi.

The Attacker's View (Open Wi-Fi)



The Evolution of Wi-Fi Encryption

Protocol	Cipher	Why It Was Replaced	Status
WEP (1997)	RC4 (24-bit IV)	IVs repeat after ∼5,000 packets; cracked in minutes by replaying captured traffic.	**Dead. Never use.**
WPA (2003)	TKIP (RC4 wrapper)	A band-aid to run on old WEP hardware; TKIP is still RC4-based and was later broken.	**Deprecated.**
WPA2 (2004)	AES-CCMP (128-bit)	Dictionary attacks possible against the 4-way handshake if the passphrase is weak.	**Widely used.** Strong with a long passphrase.
WPA3 (2018)	AES-GCMP-256 + SAE	None widely exploited.	**Current standard.** Mandatory on 6 GHz.

WPA3's Key Improvement: Forward Secrecy via SAE

WPA2 uses a fixed pre-shared key to derive session keys. If an attacker records your encrypted traffic *today* and learns your password *later*, they can decrypt everything.

WPA3 uses **SAE (Simultaneous Authentication of Equals)**, which generates a fresh, unique session key for every connection. Even if the password is stolen later, old captured traffic cannot be decrypted.

Personal vs. Enterprise Authentication

WPA2/3-Personal (Pre-Shared Key)

- Everyone on the network types the *same* passphrase.
- Easy to set up; fine for homes and small businesses.
- **Critical Flaw for Enterprises:** When an employee is fired or a device is stolen, you must change the Wi-Fi password on *every single device*. There is no way to revoke one person without affecting everyone.

WPA2/3-Enterprise (802.1X / RADIUS)

- Each user authenticates with their **own unique credentials** (username/password or digital certificate).
- The AP talks to a **RADIUS server**, which checks credentials against Active Directory.
- **Revocation is instant:** Disable one AD account and that employee's Wi-Fi disappears—nobody else is affected.
- Required for any organization with more than a handful of employees.

Hotel Key Card Analogy

PSK is like one master key that opens every room. If it's lost, you re-key the entire hotel. 802.1X is like individual key cards—deactivate one card without affecting anyone else.

Managing Guest Access: BYOD and Captive Portals

- **BYOD (Bring Your Own Device):** Employees use personal phones and laptops on the corporate network. IT does not own or control these devices—they are inherently untrusted.
- **BYOD Best Practice:** Place personal devices on an isolated VLAN (e.g., VLAN 50 "Corp-BYOD") with internet access but no access to internal servers or file shares.
- **MDM (Mobile Device Management):** Software pushed to personal devices that enforces security policies—requiring a PIN, encrypting storage, and allowing remote wipe.

Captive Portals

A captive portal intercepts all web traffic from a new client and redirects to a *landing page*—typically to accept Terms of Service, pay for access, or enter an email. Used in hotels, airports, and coffee shops.

Captive Portal Warning

Captive portal networks are almost always **open** (no Layer 2 encryption). Even after you "agree" to the terms, anyone nearby can still sniff your unencrypted traffic. Always use HTTPS and/or a VPN on these networks.

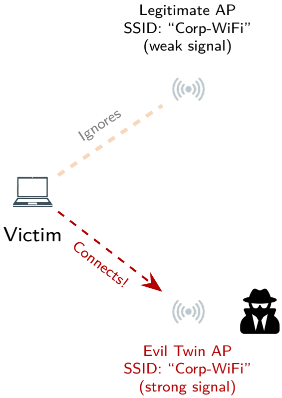
Wireless Threats: Rogue APs and Evil Twins

Rogue Access Point

- An *unauthorized* AP connected to the corporate network—usually by an employee who wants better Wi-Fi at their desk.
- **Danger:** A rogue AP bypasses the firewall, RADIUS server, and WLC. Anyone who connects is inside the network with no security checks.
- **Detection:** A Wireless IDS/IPS (**WIDS/WIPS**) scans for unknown BSSIDs.

Evil Twin Attack

An attacker sets up an AP with the *same SSID* as a legitimate network and pumps out a stronger signal. Nearby devices automatically roam to the stronger signal—landing in the attacker's control.



Defense

Use **802.1X**: the phone challenges the AP to present a

Case Study: Steve's Evil Twin

▶ Case Study: The Stolen Credentials

Steve Harrington, security analyst at Starcourt Mall, notices something wrong. The mall's guest network is named "Starcourt_Guest" and uses an open captive portal—visitors browse to any website and get a login page asking for their email.

Steve reviews WIDS alerts and spots a second AP broadcasting "Starcourt_Guest" from a MAC address that does not belong to any corporate AP. Looking outside, he notices a van near the food court entrance pumping a much stronger signal on the same SSID. Guest phones are automatically connecting to the van instead of the legitimate AP.

Review Questions

1. What specific type of attack is the van executing?
2. Why do the guest phones connect to the van automatically, rather than the real AP?
3. What authentication change would prevent this attack by requiring both sides to prove identity?

Case Study Solution: Steve's Evil Twin

✓ Solution: The Stolen Credentials

1. **Evil Twin Attack:** The attacker is spoofing the legitimate SSID with a high-power AP, serving a fake captive portal to harvest email addresses—or worse, credentials.
2. **Automatic Roaming to Strongest Signal:** Wi-Fi client drivers prefer the AP with the highest received signal strength (RSSI). Because the van's AP blasts higher power, phones associate to it automatically—there is no user prompt.
3. **802.1X with Mutual Authentication:** If the network used 802.1X (PEAP or EAP-TLS), the exchange requires the *AP* to present a valid digital certificate from a trusted CA, and the client verifies it. The van's AP has no valid corporate certificate—the phone silently fails authentication and refuses to connect, even though the signal is strong.

Key Takeaway

An open captive portal is trivial to clone. 802.1X prevents evil twin attacks because you cannot fake a certificate—the evil twin is exposed the moment the client asks "prove who you are."

Section 5: Wireless Troubleshooting

The Physical Environment: Your Signal's Worst Enemy

- Unlike a cable, an RF signal must travel through walls, floors, furniture, and people. Every obstacle weakens the signal (**attenuation**).
- **Absorption:** Dense materials convert RF energy into heat. Concrete, brick, and even human bodies absorb 2.4 GHz aggressively.
- **Reflection:** Metal objects—filing cabinets, elevator shafts, chain-link fences—bounce the signal. Reflected copies arrive out of phase and cause **multipath interference**.
- **Refraction:** Water and glass can bend the wave's path.
- **Rule of Thumb:** The 5 GHz band is attenuated *more* by walls than 2.4 GHz, because of its shorter wavelength.

Obstacle	Typical Loss
Open air	Low (inverse square)
Drywall / plywood	Low (~3 dB)
Glass window	Low–Medium (~3 dB)
Brick / cinderblock	High (~10 dB)
Concrete floor/wall	Very High (~15 dB)
Metal / elevator shaft	Signal-blocking
Human body	~3–5 dB at 2.4 GHz

The dB Scale (Rule of 3s and 10s)

- -3 dB $=$ signal power cut in half
- $+3$ dB $=$ signal power doubled
- -10 dB $=$ signal power reduced to $\frac{1}{10}$th

Signal Metrics: RSSI and SNR

- **RSSI (Received Signal Strength Indicator):** How loudly your device hears the AP. Measured in *negative* dBm—closer to zero = stronger. A value of −65 dBm is excellent; −90 dBm means the device can barely hear the AP.
- **Noise Floor:** The constant background hum of RF energy—neighboring networks, microwaves, cosmic radiation. Typically around −90 to −100 dBm.
- **SNR (Signal-to-Noise Ratio):** The difference between RSSI and noise floor. If signal is −65 dBm and noise is −90 dBm, SNR = 25 dB—excellent.
- **Why SNR matters more than RSSI:** A signal of −70 dBm in a noisy room (noise at −72 dBm) is *unusable*. The same −70 dBm in a quiet room (noise at −95 dBm) is perfectly fine.

RSSI	Performance
−30 dBm	Perfect (next to AP)
−65 dBm	Excellent (enterprise standard)
−70 dBm	Good (reliable delivery)
−80 dBm	Poor (dropped packets)
−90 dBm	Unusable (disconnects)

SNR Calculation Example

Signal: −65 dBm
Noise Floor: −90 dBm
SNR = 25 dB (Excellent)

SNR $<$ 10 dB: cannot decode the signal.
SNR $>$ 20 dB: clean, fast connection.

Non-Wi-Fi Interference: Invisible Channel Blockers

- The 2.4 GHz frequency is an **ISM band** (Industrial, Scientific, Medical)—license-free frequency shared by many devices.
- Non-Wi-Fi transmitters do not use CSMA/CA. They transmit whenever they want, generating RF noise that *prevents* Wi-Fi devices from seeing a clear channel.

CSMA/CA Under Attack

When a microwave oven is running, it floods 2.4 GHz with noise. Every Wi-Fi device's CCA check says "channel busy." Devices wait indefinitely (high latency). When they finally try to transmit, noise corrupts the frame and no ACK arrives—causing retransmits. The channel effectively freezes.

Source	Affected Band
Microwave oven	2.4 GHz
Baby monitor (analog)	2.4 GHz
Bluetooth devices	2.4 GHz
Cordless phone (older)	2.4 GHz
Radar / TDWR	5 GHz (DFS)
Neighboring Wi-Fi	2.4 or 5 GHz

Mitigation

- Move critical devices to 5 GHz.
- Use a spectrum analyzer to identify the noise source.
- Replace old/leaky equipment.

Common Configuration and Placement Mistakes

Physical Placement Errors

- **Too high:** An AP on a 30-foot warehouse ceiling is too far from floor-level devices. Signal path loss accumulates over vertical distance.
- **Above ceiling tiles:** Placing APs above a drop ceiling next to metal HVAC ducts degrades and reflects the signal before it reaches the floor.
- **Wrong orientation:** Mounting an omnidirectional AP sideways on a wall directs half the donut-shaped signal into the floor or ceiling.

Logical Configuration Errors

- **Bad channel assignment:** Assigning a 2.4 GHz AP to Channel 3 or 4 creates adjacent-channel interference with every neighboring AP on Ch 1 or 6.
- **Capacity overload:** 100 students in an auditorium, all on one AP. Wi-Fi is shared—airtime is exhausted even if the signal is strong.
- **Frequency mismatch:** A 2.4 GHz-only IoT thermostat cannot see an SSID broadcasting only on 5 GHz. Always broadcast on both bands or create a dedicated 2.4 GHz SSID for legacy devices.

Case Study: Hopper's Microwave Mystery

▶ Case Study: The Lunchtime Latency

Jim Hopper is IT Director for the Hawkins Police Department. Every weekday at **exactly 12:30 PM**, the dispatch floor erupts with complaints: wireless VoIP phones drop calls, and the CAD (Computer Aided Dispatch) software loses its Wi-Fi connection entirely.

Hopper checks the WLC and sees the dispatch AP reporting **90% channel utilization** and hundreds of packet retries—despite only 5 connected devices. The disruption lasts about 10 minutes and then completely resolves itself. No configuration changes have been made.

Review Questions

1. What is the most likely cause of a consistent, time-based RF disruption that resolves in under 15 minutes?
2. Mechanically, why does this non-Wi-Fi source cause packet retries and high channel utilization rather than just signal loss?
3. Propose two solutions—one physical, one logical (network-layer).

Case Study Solution: Hopper's Microwave Mystery

✓ Solution: The Lunchtime Latency

1. **A Leaky Microwave Oven:** Someone in the break room heats lunch at 12:30 PM. Microwave ovens operate at 2.4 GHz—the same band as the dispatch floor's Wi-Fi. An older or damaged microwave can radiate significant RF energy outside its shielded enclosure.
2. **CSMA/CA Paralysis:** Microwaves do not respect CSMA/CA. Wi-Fi devices' CCA checks detect the channel as "busy" and defer indefinitely (causing high latency). When a device tries to transmit anyway, the microwave noise corrupts the frame, the ACK is never received, and the device retransmits—causing high retry counts and channel utilization.
3. **Two Solutions:**
   - *Physical:* Replace the old microwave with a properly shielded modern unit.
   - *Logical:* Move the dispatch APs and devices to the **5 GHz band**, which is not affected by 2.4 GHz microwave ovens.

Module 12 Summary

Sections 1–2: RF Basics and Standards

- 2.4 GHz = range; 5 GHz = speed; 6 GHz = clean, WPA3-required.
- Use only channels **1, 6, 11** on 2.4 GHz to avoid ACI.
- CSMA/CA (listen, wait, transmit, ACK); RTS/CTS solves hidden node.
- b (DSSS) → n (MIMO) → ac (MU-MIMO) → ax (OFDMA) → be (MLO).

Section 3: Enterprise Design

- Each SSID maps to a VLAN; AP uplink must be a trunk.
- Thin APs + WLC (CAPWAP) enable centralized management.
- Site survey before installation; microcells over high power for density.
- Omni for open offices; directional for hallways and signal containment.

Section 4: Security

- WEP is dead; WPA2 (AES) is adequate; WPA3 (SAE) is the standard.
- Personal (PSK): shared password—fine for homes only.
- Enterprise (802.1X): per-user credentials via RADIUS—required for business.
- Rogue APs bypass security; evil twins harvest credentials.
- 802.1X + mutual auth defeats evil twins.

Section 5: Troubleshooting

- Target RSSI ≤ -65 dBm and SNR ≥ 20 dB.
- Non-Wi-Fi ISM interference (microwaves, Bluetooth) paralyzes CSMA/CA.
- Fix: move to 5 GHz, fix placement, use only 1/6/11 on 2.4 GHz.